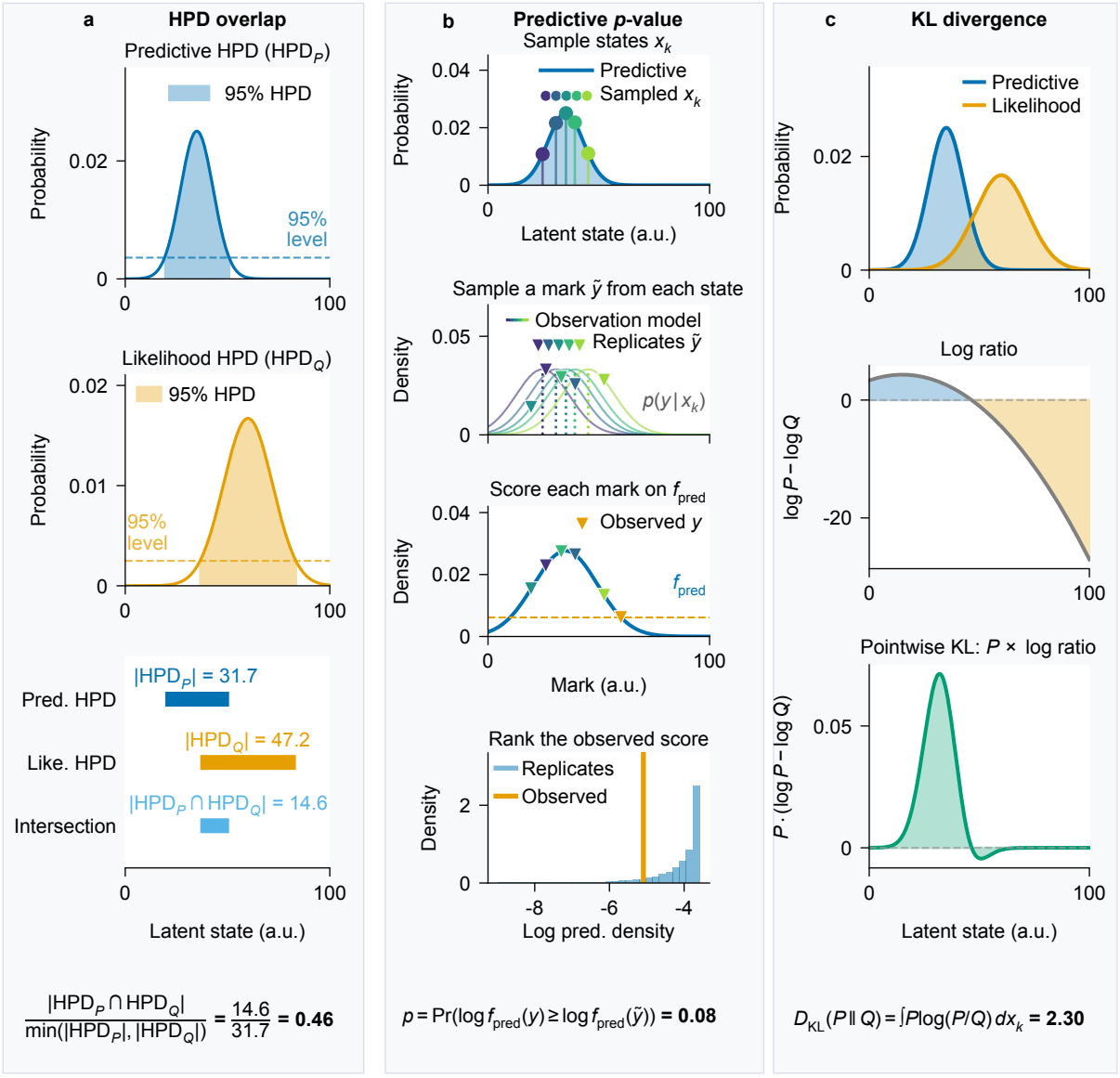

a
HPD overlap
Predictive HPD (HPD_P)
95% HPD
Probability
0.02
0
95%
level
0
100
Likelihood HPD (HPD_Q)
95% HPD
0.02
0.01
95%
level
$|HPD_P|$ = 31.7
Pred. HPD
$|HPD_Q|$ = 47.2
Like. HPD
$|HPD_P \cap HPD_Q|$ = 14.6
Intersection
Latent state (a.u.)
$\frac{|HPD_P \cap HPD_Q|}{\min(|HPD_P|, |HPD_Q|)} = \frac{14.6}{31.7}$ = **0.46**
b
Predictive *p*-value
Sample states x_k
Predictive
Sampled x_k
0.04
0.02
Latent state (a.u.)
Sample a mark $\tilde{y}$ from each state
Observation model
Replicates $\tilde{y}$
Density
0.05
$p(y \mid x_k)$
Score each mark on f_{pred}
Observed y
f_{pred}
Mark (a.u.)
Rank the observed score
Replicates
Observed
2
-8
-6
-4
Log pred. density
$p = \Pr(\log f_{\text{pred}}(y) \geq \log f_{\text{pred}}(\tilde{y}))$ = **0.08**
c
KL divergence
Predictive
Likelihood
Log ratio
$\log P - \log Q$
-20
Pointwise KL: $P \times$ log ratio
$P \cdot (\log P - \log Q)$
0.05
Latent state (a.u.)
$D_{\text{KL}}(P \| Q) = \int P \log(P/Q)\, dx_k$ = **2.30**